USSCOS Build Status and Priorities

What is ready, what is close, what remains · 28 September 2026

Summary. The sales side of USSCOS is built and working — quoting, orders, invoicing, customers, products, reps, artwork. The warehouse side was not: of the seven ways stock moves at USSC, one and a half were built.

All seven are now built. Receiving, shipment, purchase-order receipts, transfers, returns, adjustments and counting — every one through a single path that writes the movement and the per-location balance together, inside one database transaction. The product total is recomputed from its locations rather than incremented, so it cannot drift from the detail.

Shipping is built too — pack verification, FedEx labels with the tracking number captured automatically, and an invoice review queue so the customer is told only once the invoice is finished.

What is left is mostly not software. The screens exist; what they now need is people to walk the buildings and count, a product file with weights and barcodes, and the process signed off by those who work it.

Accounting is not on the critical path. QuickBooks keeps the books, bills stay in QuickBooks, and none of it blocks going live.

Completed since the last version of this document

All deployed and tested against the live database.

Item	What it does now
Receiving	Scan or type a SKU, confirm the quantity, choose the location. Suggests a quantity from the pack size but never insists on it
Deduct stock on shipment	Shipping takes stock out of the location holding it, biggest holding first, splitting across locations when needed
Stock on every invoice	A counter sale or a correction moves stock too, not only a shipment. Editing an invoice moves the difference
Receive against a PO	Records what actually arrived, never trimmed to match the PO, into a location
Variance queue	Receipts that disagree with their PO, resolved by updating the PO or explaining the difference
Locations	Admin screen, warehouses named 1000, 900, 3085 and 730
Customer PO on the job	The PO document itself is filed on the job binder and stays with the job after invoicing
Pack quantities	Aerosol, Fat Cans, pails and 2.5 gal jugs — 508 of 885 products packed, 458 weighed

Item	What it does now
Sales tax engine	Tax calculated here, from the ship-to address. Rate, county, base and reason frozen onto each invoice
GA and NC rate tables	All 259 counties and 1,605 ZIPs, from the state revenue departments
Tax reporting	Liability by county, Amazon's collections kept separate, and a sales-by-state nexus watch
Adjustments	The honest correction path, with a required reason code, plus the negative-stock worklist
Transfers	Scanned out of one building and into the other. In between it counts at neither, which is correct
Returns	Only resellable goods go back on the shelf. Credit priced from the original invoice and its frozen tax rate
Cycle counting	Blind counts — the expected figure is never sent to the counter — reviewed, then applied as adjustments
Pack verification	A second scan at the bench, checked against the pick rather than the order. Refuses anything not on the order
Shipment tracking	Several numbers per shipment, each a carrier link, searchable, and kept on the invoice
Invoice review queue	A shipped invoice waits for handling to be added. Approving it is what emails the customer
FedEx	Labels created and the tracking number captured with nobody retyping it. Proven against sandbox
Box rules	One aerosol case is a single box, two go in a double, three go as a double plus a single. One pail is a parcel, two are freight
Dashboard	Real figures, and only the ones USSCOS can actually know. The queues from every module surface here
Credit memos	Raised from a return. Negative totals, so every existing report nets them correctly
Job binder	Artwork, the customer PO, append-only job notes, and a quality checklist USSC defines itself
Box dimensions	All four boxes measured — aerosol single and double, Fat Can, Robo jug — with a 2 lb empty weight each
Rate cross-check	Every GA and NC ZIP compared against an independent published table. Ambiguity is now found rather than assumed
Rate refresh	One command loads whatever chart is dropped in, dates it from inside the chart, and the dashboard warns before rates go stale

The design point underneath all of it. Every stock movement goes through one function that writes the transaction record and the per-location balance inside a single database transaction. The product total is *recomputed* from the sum of its locations rather than incremented, so it cannot drift away from the detail. That is the difference between a count that can be trusted and the one being replaced.

Problems found and fixed on the way

Each of these was live and silent. They are listed because they may have already cost money, and because they say something about where else to look.

What was wrong	What it cost
Shipping the rest of a part-shipped order re-invoiced everything picked	The customer was billed a second time for the first shipment. Worth checking against real partial shipments in QuickBooks
Receiving against a PO capped the quantity at what was ordered	620 cases arriving against a PO for 600 silently became 600 — twenty cases gone, no error, no note
PO receipts never touched locations	Stock received against a PO existed as a total in no building, invisible to the location screens
Invoices could be edited without stock following	Correcting an invoice from ten to eight billed eight and left ten deducted
A database error on an invoice showed as “Not Found”	Real faults looked like missing records, so nobody would have reported them as faults
North Carolina was taxed at 8.25 percent statewide	That is Mecklenburg's rate. Most NC counties are 6.75 or 7 percent, so customers would have been overcharged — caught by an Amazon packing slip
Every Georgia ZIP pointed at a rate not yet in effect	Caught in my own work before anyone used it. It would have quietly started working on 1 October, which is the worst way for a fault to behave
One county stood in for a whole state	With no default marked, the lowest-numbered county — Appling, 8 percent — was returned as the rate for the whole of Georgia
ZIPs were assigned to counties by land area	A weak proxy: a ZIP can hold most of its acreage in a rural county and nearly all its mail in the urban one next door, which priced Johns Creek as Gwinnett. Checking against a published table found 281 such ZIPs
Georgia's October rates had no end date	They would have gone on applying silently long after the quarter ended. Found while automating the refresh, which now judges rates by the age of the newest chart rather than trusting an end date
Fat Cans were sharing the aerosol box	Every 12-to-a-case product had been given the same box, so 26 oz cases would have been rated as the smaller 18 oz carton. They have their own now
A Robo jug pack was recorded as the box, not the jug	An order for three jugs would have been weighed as three pairs — a FedEx label reading 105 lb for something that weighs 35
Nothing told a credit memo from an invoice	invoice_type has a credit_memo value that nothing filtered on, so a credit would have counted as revenue in Sales by Rep, A/R aging and customer lifetime value. Fixed by giving credits negative totals, so all fifteen money queries net correctly untouched

What was wrong	What it cost
Shipping queued invoices from the wrong place	The rule lived in one controller, so anything else that shipped an order would have skipped the review queue and never told the customer
FedEx sandbox answers every address check with a fake	It placed 1000 McFarland 400 Blvd in Chile. Address validation now reports that it proved nothing, rather than a verdict nobody should believe

Ready to use now

Area	What it does
Leads and pipeline	Web forms feed leads, routed by type, converted to customers
Quotes	Built, emailed, converted to orders; status flows back to the lead
Sales orders	Created, edited, packing slips, payment collection
Invoicing	Ship and invoice, receipt or invoice email, closes the order, moves stock
Customers	41,233 records, 360° profile, revenue history, activity timeline, alerts
Duplicate prevention	Adding a customer warns on likely duplicates, including ones the user cannot see
Products	885 active records, images, documents, ~100 fields for EDI and Amazon
Sales reps	33 reps, credited revenue, commission rates, logins
Reporting	Sales by rep, sales by employee, A/R aging, all with date ranges
Pick and verify	Scan against the order, refuse wrong items, report short stock
Receiving	Scan in, suggested pack quantities, into a named location
Stock movement	Receipts, shipments, invoice corrections and PO receipts, all logged by location
Purchasing	POs raised and received against, with a variance queue when they disagree
Locations	Warehouses 1000, 900, 3085, 730, with bays and racks addable
Job binder	Artwork with revision history and approvals, plus the customer's PO filed on the job
Access control	Reps and distributors restricted to their own customers, enforced in the data layer
Admin	Users, reps, terms, carriers, tax rates, locations, with safe deactivate and delete
Backups	Nightly, verified by restoring, copied off-site to DigitalOcean

Close to finished

Item	What remains	Effort
Pack and weight data	508 products packed and 458 weighed. The rest wait on figures, and all of it must survive the reimport	Small
Locations	Screens and warehouses done. Needs the bays and racks in 730 entering	Small
FedEx production	Works against sandbox. FedEx must certify the Ship API before real labels print, and package weights are still missing	Small
Sales tax	Engine and both states' rates are in. Needs real invoices checked against what QuickBooks charged before it is trusted	Small
Quality checklist	The screen is built and deliberately empty — USSC adds the checks it actually performs	Operations

Left to build

Grouped by what they belong to. Effort is relative, not a quote.

Inventory — still the critical path

Item	Why it matters	Effort
Opening physical count	One full count. The screens exist and a full count is the same machinery as a cycle count — this is people walking buildings, not code	Operations
Bays and racks in 730	Locations work, but 730 has no internal detail yet, so everything there counts as one place	Operations
Scanners	None bought yet, so picking, packing and counting are keyboard-only. 738 products carry no barcode either — printing Code 128 SKU labels would settle both	Operations

Shipping and carriers

Item	Why it matters	Effort
Freight section and BOL	Carrier, cost, pallets and a generated bill of lading. Freight PRO numbers are hand-entered today	Medium
Automatic handling fee	Calculated rather than typed, so the review queue becomes a one-click check instead of data entry	Small
Tracking status	Delivered or in transit on the invoice. The second FedEx project is created and its keys are in	Small
Kuebix quotes	Removes retyping. Pending confirmation the API is enabled	Medium
Amazon orders direct	Straight into USSCOS as same-day priority, instead of being keyed by hand	Medium

Data and integration

Item	Why it matters	Effort
Product reimport	765 SKUs with barcodes and pack quantities replacing the current 885	Small
Carry pack data through the reimport	The figures entered so far must survive it or they are lost	Small
Quarterly tax rate refresh	Loading is now one command and the dashboard warns when rates go stale. What cannot be automated is the download — Georgia's site refuses our server and Avalara's tables sit behind a form. Two minutes, four times a year	Operations
Historical A/R	Imported invoices carry no payments, so $1.6m reads as overdue that is not. Either import the payments or close the period	Small
281 disputed ZIPs	Where our county assignment and the published table disagree — 174 in GA, 107 in NC. Listed on the tax screen and flagged for a person on each order. Settling them properly means address-level lookup	Medium

Item	Why it matters	Effort
QuickBooks bridge	Reads QuickBooks and talks to USSCOS. Outbound only — no port to open	Large
Capture QuickBooks IDs	So a rename in QuickBooks never breaks the link again	Small
Two-entity support	TCC and USSC, with per-area access rather than one flag	Large

Later — genuinely optional for go-live

Customer and distributor portals with reordering · tiered and volume pricing · rep dashboard and commission statements · public website · two-factor authentication and audit log · Mar-Kov integration · accounting ledger, if it is ever wanted.

Waiting on other people

Not stalled for want of effort — listed so they are not mistaken for work in progress.

Item	Waiting on
Remaining pack quantities and weights	Roughly 380 products still without a pack size or a weight. Aerosol, Fat Cans, pails and 2.5 gal jugs are complete
Robo jug boxing decision	Settled for parcels — one jug per box, since a pair is 58 to 70 lb. The box still holds two on a pallet or a truck, and the pallet is 48 jugs either way
Process sign-off	The step-by-step handed to sales, shipping, bookkeeping and operations to approve or mark up
Pail dimensions	A pail ships unboxed, so the pail is the package and FedEx needs its size. The four cardboard boxes are all measured
Scanners	None bought yet. Two to trial would do — one at receiving, one at the packing bench
Final product file	Barcodes, QuickBooks IDs, case weights, and the pack columns
Hazmat class	Aerosol is regulated and FedEx will refuse production labels without it declared
FedEx certification	FedEx must certify the Ship API before production labels can print
SKU renaming	Being finalised in QuickBooks
Kuebix automation	Confirmation the API is enabled on the subscription, and its cost
FedEx	Account number and API credentials
Operations walkthrough	The interview, which settles who owns stock at the canning company
Inventory valuation	The accountant. Blocks costing stock, not counting it
Invoice fields (terms, ship via, PO)	A QuickBooks export carrying those columns
Distributor list	Turf Tank, BSN and the others

The order I would build it in

> **The principle.** Every way stock moves needs a screen, and each screen must be quicker than not using it. Any movement without one becomes the place where someone shrugs and the count starts rotting again — which is exactly how the current numbers went wrong.

	Phase	Contains	State
1	**Stock can move**	Receiving · deduct on shipment · PO receipts · transfers · adjustments	Done
2	**Stock is right**	Returns · cycle counting · negative stock list · opening physical count	Built — the count itself is operations

	Phase	Contains	State
3	**Product data**	Reimport 765 SKUs with barcodes, pack quantities, QuickBooks IDs	Waiting on the file
4	**Shipping finished**	Pack verification · PO receiving · freight and BOL · handling fee	Built except freight — paused for sign-off
5	**Carriers**	FedEx labels and tracking · Kuebix quotes · Amazon direct	FedEx proven in sandbox
6	**QuickBooks**	The bridge, both files, items and customers pulled	Not started
6b	**Tax upkeep**	Quarterly GA rate refresh · the 281 disputed ZIPs	Loading automated, downloading is manual
7	**Two entities**	TCC and USSC with per-area access	Not started
8	**The rest**	Dashboard · job binder remainder · portals · tiered pricing	Not started

Phases 1 to 4 are what "functional" means. Five onwards make it faster and wider, but the warehouse can run without them.

> **One sequencing point worth protecting.** The physical count belongs in phase 2, *after* the movement screens exist. Counting first means the count is out of date before anyone can use it, and the second count is much harder to ask for than the first.

Questions that need answers

Grouped by who can answer them. The starred ones change what gets built.

Answered since the last version

- **How does a pallet count?** As a quantity of base units, not as its own item — 108 cases × 12 = 1,296 cans for 18 oz aerosol, 75 × 12 = 900 for Fat Cans, 24 pails, 48 jugs. Built that way.
- **Is the PO to TCC tied to a job?** No. It is replenishment, billed to USSC, and never belongs to a particular sales order.
- **Which PO belongs on the job binder?** The customer's, as a document. Built.
- **Should we use Stripe for tax lookups?** No — priced per calculation and built to sit under Stripe payments. Two states is a rate table we own, and it is built.
- **Do we charge tax outside GA and NC on web sales?** No, only where we have nexus. Sales by state are now tracked so a threshold is seen coming.
- **Should Amazon's tax be stored?** Yes, as a memo figure that never touches what we owe. Built.

Operations

- ★ **While paint sits at the canning company, is it ours or TCC's?** Decides whose inventory it is.
- ★ **Who watches the variance queue?** It is built, but a queue nobody opens is the same as no queue.
- ★ **When is the opening count, and who does it?** Everything else is ready for it. Until it happens the figures are a starting point, not a count.
- How often should a cycle count run, and over what — highest value, fastest moving, or simply everything in turn?
- How much paint is lost turning totes into cans? Nobody tracks it, but someone knows.
- What bays and racks exist in 730, and how are they labeled?
- Is the draft-to-confirmed step happening on every order, or are some invisible to shipping?
- When is stock considered gone — at pick, at load, or at invoice? It currently comes out at invoice.
- Who prints labels, and on what printer?

Bookkeeper and accountant

- ★ **Who owns inventory valuation — USSCOS or QuickBooks?** Blocks costing stock, though not counting it.
- ★ **Do our rates match QuickBooks?** Check a handful of real GA and NC invoices against what QuickBooks charged. The tables come from the states, but that is the comparison that proves it.
- **Have any part-shipped orders been billed twice?** The fault is fixed, but any that already happened are in QuickBooks, not here.
- **Where did the 8.25 percent NC rate come from?** It is Mecklenburg's. Worth knowing whether it was ever charged outside USSCOS.
- After the rename, do historical invoices show the new item names in an export?
- Can the export include the QuickBooks ListID column?
- How is sales tax set up — one rate per state, or county codes?
- Are processor fees captured per transaction or reconciled monthly?

Suppliers and vendors

- ★ **Kuebix: is API access enabled on our subscription, and does it cost extra?**
- FedEx: account number and API credentials.
- Mar-Kov: is there an API, and is its QuickBooks link switched on today?

Management

- ★ **Will TCC staff use USSCOS at go-live?** Two-entity support is far cheaper to add during the data reimport than afterwards.
- Does anyone need a combined TCC and USSC view? Neither QuickBooks file sees the other, so only USSCOS could produce one.
- Who is the pilot group for the scanners, and when can they spend a day on it?

> **The single most useful thing right now** is still the finished product file — barcodes, QuickBooks IDs and the remaining pack quantities. Scanning, transfers and counting all wait on it, and no amount of building substitutes for it. The pack figures supplied so far cover 508 of 885 products and must be carried through the reimport, or they are lost.